Individual Report

MiniOS Alpha | Operating Systems Final Project

Student Name:

Group: Marquez and Friends

Assigned Role: User Management + System Information + Testing

What did I contribute to the project?

I built the login system in users.py and the systeminfo screen, and I was responsible for testing the whole program. I ran the tests for every required command and the error cases from the project sheet, and I recorded the results in the Technical Documentation.

What features did I personally develop?

- A UserManager class with two accounts: admin (Administrator) and student (Standard User)
- Password hashing with SHA-256, and a login screen with hidden password entry and three attempts
- whoami (required), plus users and logout
- systeminfo with the OS name, version, architecture, virtual CPU, memory, user and role, and number of running processes
- The test sessions and the results table

What problems did I encounter?

- The password prompt showed a warning and echoed the typing when the input was not a normal console.
- Passwords should not be stored as readable text.
- After logout, processes from the previous user could still be running.
- My first test run gave wrong results because each word of the test file was on its own line.

How did I solve those problems?

- I use getpass and fall back to plain input if it cannot hide the typing. In a normal Command Prompt the password stays hidden.
- I store only SHA-256 hashes and compare hashes at login.
- logout calls kill_all so memory and processes are cleaned up before the next login.
- I rewrote the test file with one full command per line and re-ran everything, which showed the shell had been working all along.

What did I learn from this project?

I learned how authentication works and why passwords are hashed. I also learned that testing has to cover wrong input and edge cases, not only normal use, and that a badly built test can look like a program bug. Writing a repeatable test session saved a lot of time after each fix.

Evidence of my contribution:

- users.py (UserManager)
- The systeminfo, whoami, users and logout commands in main.py
- The Testing section and results table in Technical_Documentation.pdf

Sample output from my part of the program:

```
Username: student
Password: ********

Login successful.

Welcome, student!

MiniOS \> whoami
Current User: student

MiniOS \> systeminfo
OS Name: MiniOS Alpha
Version: 1.0
Architecture: 64-bit
CPU: Virtual CPU
Memory: 1024 MB
User: student (Standard User)
```

Screenshots (attach in the screenshots folder and list the file names here):